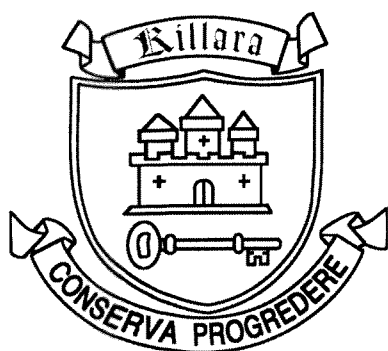




**KILLARA HIGH SCHOOL
MATHEMATICS
Extension 2
Assessment Task
March 2008**

TASK WEIGHTING:

30 %

NAME: _____

Time allowed: 120 short minutes plus 5 minute reading time

The following outcomes will be assessed in this task

H1-9, HE6, HE7, E2, E3, E4, E6, E8

Instructions:

- Show all necessary working
- All questions may be attempted
- Marks may be deducted for careless or poorly arranged work
- Only approved calculators may be used
- Label each question carefully

This is a School Based paper only. The content does not necessarily reflect that of the Higher School Certificate Examination paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1: x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \log_e x, x > 0.$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \log_e \{x + \sqrt{x^2 - a^2}\}, |x| > |a|$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \log_e \{x + \sqrt{x^2 + a^2}\}$$

Question 1. (15 marks) Complex Numbers**Start a new page**(a) Given $z = \sqrt{6} - \sqrt{2}i$, find:

(i) $\operatorname{Re}(z^2)$; [1]

(ii) $|z|$; [1]

(iii) $\arg z$; [2]

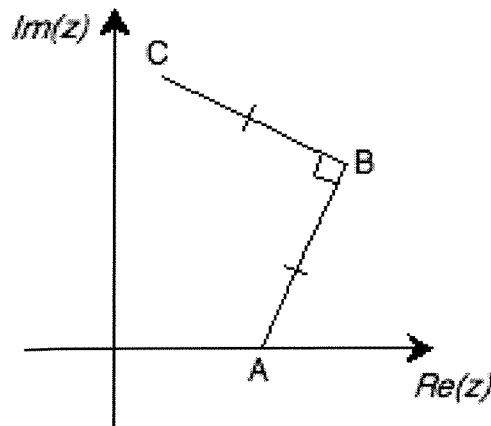
(iv) z^4 in the form $x + iy$. [2]

(b) The equations $|z - 8 - 6i| = 2\sqrt{10}$ and $\arg z = \tan^{-1} 2$ both represent loci on the Argand plane.(i) Write down the Cartesian equations of the loci, and hence show that the points of intersection of the loci are $2 + 4i$ and $6 + 12i$. [3]

(ii) Sketch both loci on the same diagram, showing their points of intersection.

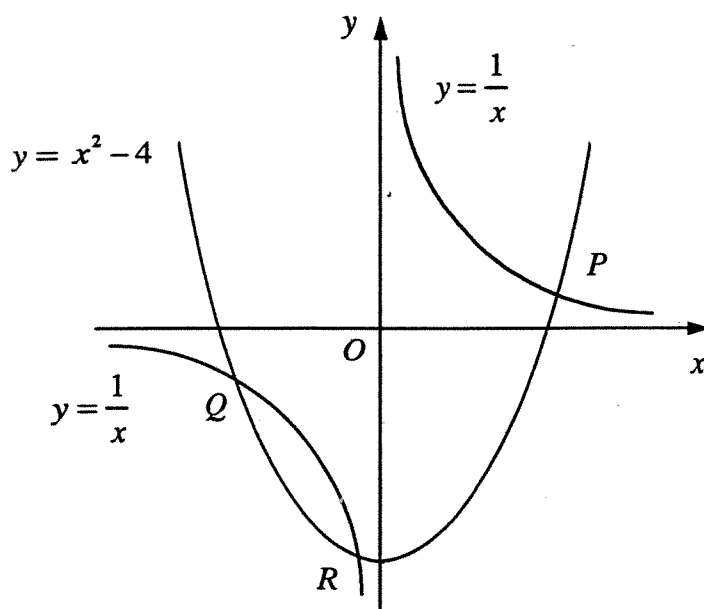
(You need not show the intercepts with the axes.) [2]

(c)

The diagram above shows the fixed points A , B and C in the Argand plane, where $AB = BC$, $\angle ABC = \frac{\pi}{2}$, and A , B and C are in anticlockwise order. The point A represents the complexnumber $z_1 = 2$ and the point B represents the complex number $z_2 = 3 + \sqrt{5}i$.(i) Find the complex number z_3 represented by the point C . [2](ii) D is the point on the Argand plane such that $ABCD$ is a square. Find the complex number z_4 represented by D . [2]

Question 2. (20 marks) Polynomials**Start a new page**

(a)



The curves $y = x^2 - 4$ and $y = \frac{1}{x}$ intersect at the points P , Q and R where $x = \alpha$, $x = \beta$ and $x = \gamma$.

(i) Show that α , β and γ are the roots of the equation $x^3 - 4x - 1 = 0$. [2]

(ii) Find a polynomial equation with numerical coefficients which has roots α^2 , β^2 and γ^2 . [2]

(iii) Find the polynomial equation with numerical coefficients

which has roots $\frac{1}{\alpha^2}$, $\frac{1}{\beta^2}$ and $\frac{1}{\gamma^2}$ [2]

(iv) Hence find the numerical value of $OP^2 + OQ^2 + OR^2$ [2]

(b) (i) Use DeMoivre's theorem to show that:

$$(1 + i \tan \theta)^n + (1 - i \tan \theta)^n = \frac{2 \cos n\theta}{\cos^n \theta} \quad (\cos \theta \neq 0) \text{ where } n \text{ is a positive integer. [3]}$$

(ii) Hence show that the equation $(1 + z)^4 + (1 - z)^4 = 0$ has roots $\pm i \tan \frac{\pi}{8}$, $\pm i \tan \frac{3\pi}{8}$. [2]

(iii) Hence show that $\tan^2 \frac{\pi}{8} = 3 - 2\sqrt{2}$ [2]

(c) The polynomial function $P(x) = x^4 - 4x^3 - 3x^2 + 50x - 52$ has a zero at $x = 3 - 2i$.

Factorise $P(x)$ over the field of:

(i) rationals. [3]

(ii) complex numbers. [2]

Question 3. (20 marks) Conics

Start a new page

- (a) For the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$, find
- (i) the value of the eccentricity, e . [1]
 - (ii) the co-ordinates of the two foci. S and S' [2]
 - (iii) the equations of the directrices [2]
 - (iv) the equation of the tangent to the hyperbola at the point $(5, 2\frac{1}{4})$ [2]
 - (v) Make a neat sketch of the hyperbola showing these features [2]

- (b) Explain why $\frac{x^2}{k-19} + \frac{y^2}{3-k} = 1$ cannot represent the equation of an ellipse. [1]

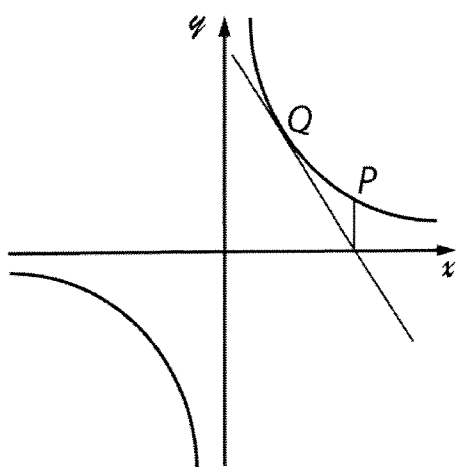
- (c) $P(a\cos\theta, b\sin\theta)$ lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

The normal at P cuts the x -axis at G and N is the foot of the perpendicular from P to the x -axis.

Show that

- (i) $OG = e^2.ON$ [3]
- (ii) $SG = e.SP$ and $S'G = e.S'P$ where S and S' are the foci. [3]

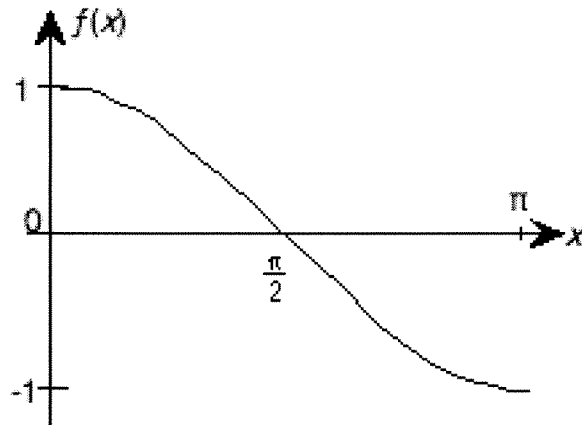
- (d) P and Q are variable points on the rectangular hyperbola $xy = c^2$.



- i) The tangent at Q passes through the foot of the ordinate (y-value) of P . If P and Q have parameters p and q , show that $p = 2q$ [2]
- ii) Hence prove that the locus of the mid-point of PQ is a rectangular hyperbola. [2]

Question 4. (20 marks) Other Sections**Start a new page**

(a)



The graph of $f(x) = \cos x$ is shown above for the interval $0 \leq x \leq \pi$.

Provide separate half-page sketches of the graphs of the following

(i) $y = [f(x)]^2$ [2]

(ii) $y^2 = f(x)$ [2]

(iii) $y = \frac{1}{|f(x)|}$ [2]

(b) (i) Resolve $\frac{5x}{(x+1)(x-2)}$ into partial fractions [2]

(ii) Hence evaluate $\int_3^4 \frac{5x}{(x+1)(x-2)} dx$ [3]

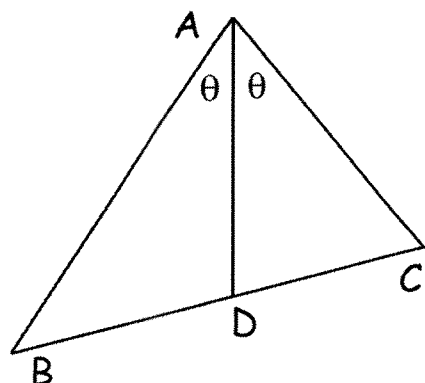
(c) (i) On the same diagram sketch the graphs of $x^2 + y^2 = 1$ and $x^2 - y^2 = 1$, showing clearly the co-ordinates of any points of intersection with the axes and the equations of any asymptotes. [4]

(ii) On a separate sketch shade the region where the inequality:

$$(x^2 + y^2 - 1)(x^2 - y^2 - 1) \leq 0 \text{ holds.} \quad [2]$$

Continued on next page →

(d)



Not to scale.

In the triangle ABC it is given that AD [3]
is the bisector of $\angle BAC$.

(i) Copy the diagram onto your paper

(ii) Prove that $\frac{AB}{BD} = \frac{AC}{DC}$

End of paper.

Question 1

a) $z = \sqrt{6} - \sqrt{2}i$

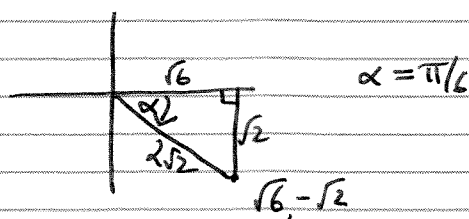
$$\begin{aligned}
 \text{(i)} \quad \operatorname{Re}(z^2) &= \operatorname{Re}(\sqrt{6} - \sqrt{2}i)^2 \\
 &= \operatorname{Re}(6 - 4\sqrt{3}i - 2) \\
 &= \operatorname{Re}(4 - 4\sqrt{3}i) \\
 &= 4
 \end{aligned}$$

[1]

$$\begin{aligned}
 \text{(ii)} \quad |z| &= \sqrt{(\sqrt{6})^2 + (\sqrt{2})^2} \\
 &= \sqrt{6+2} = \sqrt{8} \\
 &= 2\sqrt{2}
 \end{aligned}$$

[1]

$$\text{(iii)} \quad \arg z = \frac{\sqrt{2}}{\sqrt{6}} = \frac{1}{\sqrt{3}}$$



$$\arg z = -\frac{\pi}{6}$$

[2]

$$\text{iv)} \quad z^4 = ?$$

$$z = \sqrt{6} - \sqrt{2}i$$

$$z^2 = (4 - 4\sqrt{3}i) \quad \text{cf (i)}$$

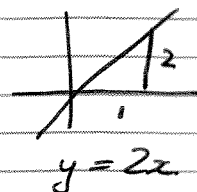
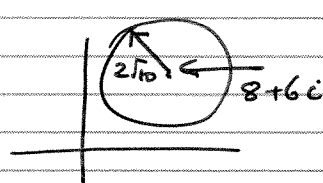
$$\begin{aligned}
 z^4 &= (4 - 4\sqrt{3}i)^2 \\
 &= 16 - 32\sqrt{3}i - 48 \\
 &= -32 - 32\sqrt{3}i \\
 &= -32(1 + \sqrt{3}i)
 \end{aligned}$$

$$\text{OR.} \quad z = \sqrt{6} - \sqrt{2}i = \sqrt{2}(\sqrt{3} - i)$$

$$\begin{aligned}
 z^4 &= [\sqrt{2}(\sqrt{3} - i)]^4 \text{ etc } [2\sqrt{2} \cos -\frac{\pi}{6}]^4 \\
 &= (2\sqrt{2})^4 \cos(-\frac{4\pi}{6}) = \dots
 \end{aligned}$$

[2]

b) $|z - 8 - 6i| = 2\sqrt{10} \quad \arg z = \tan^{-1}2$



$$(x-8)^2 + (y-6)^2 = (2\sqrt{10})^2$$

Solve for points of intersection

$$(x-8)^2 + (y-6)^2 = 40$$

$$(x-8)^2 + (2x-6)^2 = 40$$

$$x^2 - 16x + 64 + 4x^2 - 24x + 36 = 40$$

$$5x^2 - 40x + 100 = 40$$

$$5x^2 - 40x + 60 = 0$$

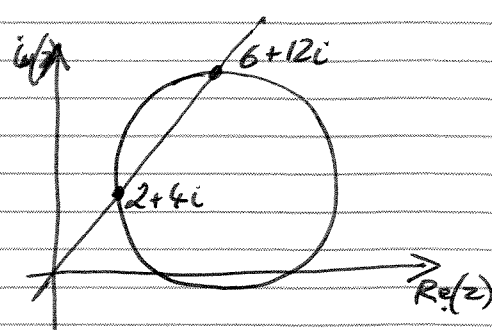
$$5(x^2 - 8x + 12) = 0$$

$$5(x-6)(x-2) = 0$$

$$x = 6, 2$$

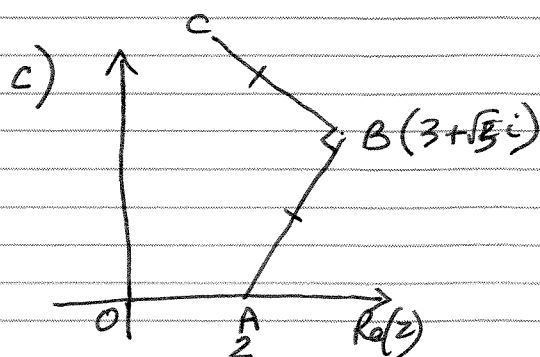
$$x = 6, y = 12 \quad (6, 12) \quad z = 6 + 12i$$

$$x = 2, y = 4 \quad (2, 4) \quad z = 2 + 4i$$



[3]

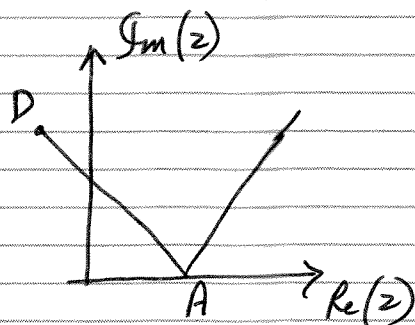
[2]



$$\begin{aligned}
 \vec{OC} &= [\vec{OA} + \vec{AB}] + \vec{BC} \\
 &= 2 + (1 + \sqrt{5}i) + i(1 + \sqrt{5}i)
 \end{aligned}$$

$$\begin{aligned}\vec{OC} &= 2 + 1 + \sqrt{5}i + i - \sqrt{5} \\ &= 3 - \sqrt{5} + (1 + \sqrt{5})i\end{aligned}$$

[2]



$$\vec{BC} = \vec{AD}$$

$$\begin{aligned}\vec{OD} &= \vec{OA} + \vec{AD} \\ &= 2 + i(1 + \sqrt{5}i) \\ &= 2 + i - \sqrt{5} \\ &= 2 - \sqrt{5} + i\end{aligned}$$

[2]

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Question 2

$$\left. \begin{aligned} \text{a) i) } y &= x^2 - 4 \\ y &= \frac{1}{x} \end{aligned} \right\} \text{ intersect at PQR}$$

$$x^2 - 4 = \frac{1}{x}$$

$$x^3 - 4x - 1 = 0 \quad \text{--- (A)}$$

Let the roots be α, β, γ [2]

$$\text{ii) } X = \alpha^2 \quad \therefore \alpha = \sqrt{X}$$

This satisfies (A)

$$(\sqrt{X})^3 - 4(\sqrt{X}) - 1 = 0$$

$$X\sqrt{X} - 4\sqrt{X} - 1 = 0$$

$$\sqrt{X}(X - 4) = 1$$

$$\left[\frac{\sqrt{X}}{X - 4} \right]^2 = 1^2$$

$$X(X - 4)^2 = 1$$

$$X(X^2 - 8X + 16) = 1 \quad \text{(A)}$$

$$X^3 - 8X^2 + 16X - 1 = 0 \quad [2]$$

(iii) for polynomial with roots $\frac{1}{\alpha^2}, \frac{1}{\beta^2}, \frac{1}{\gamma^2}$

note (A) has roots $\alpha^2, \beta^2, \gamma^2$

The polynomial we want is

$$\left(\frac{1}{X}\right)^3 - 8\left(\frac{1}{X}\right)^2 + 16\left(\frac{1}{X}\right) - 1 = 0$$

$$\frac{1}{X^3} - \frac{8}{X^2} + \frac{16}{X} - 1 = 0$$

$$1 - 8X + 16X^2 - X^3 = 0$$

$$\therefore X^3 - 16X^2 + 8X - 1 = 0 \quad \text{(B)}$$

has roots $\frac{1}{\alpha^2}, \frac{1}{\beta^2}, \frac{1}{\gamma^2}$ [2]

(iv) Say $P(\alpha, \frac{1}{\alpha}), Q(\beta, \frac{1}{\beta}), R(\gamma, \frac{1}{\gamma})$

$$OP^2 = \alpha^2 + \frac{1}{\alpha^2} \quad OQ^2 = \beta^2 + \frac{1}{\beta^2} \quad OR^2 = \gamma^2 + \frac{1}{\gamma^2}$$

$$OP^2 + OQ^2 + OR^2 = \alpha^2 + \frac{1}{\alpha^2} + \beta^2 + \frac{1}{\beta^2} + \gamma^2 + \frac{1}{\gamma^2}$$

$$= \alpha^2 + \beta^2 + \gamma^2 + \frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$$

from (A) from (B)

$$= 8 + 16$$

$$= 24$$

[2]

b/ (i)

$$(1 + i \tan \theta)^n = \left(\frac{1 + i \sin \theta}{\cos \theta} \right)^n$$

$$= \frac{(\cos \theta + i \sin \theta)^n}{\cos^n \theta}$$

$$= \frac{1}{\cos^n \theta} [\cos n\theta + i \sin n\theta]$$

by De Moivre's Theorem

Similarly

$$(1 - i \tan \theta)^n = \frac{1}{\cos^n \theta} [\cos n\theta - i \sin n\theta]$$

Q2 b1 cont

$$= \frac{1}{\cos^4 \theta} \left[\cos(-\theta) + i \sin(-\theta) \right]^4$$

$$= \frac{1}{\cos^4 \theta} \left[\cos(-\theta) + i \sin(-\theta) \right]^4$$

$$= \frac{1}{\cos^4 \theta} \left[\cos \theta - i \sin \theta \right]^4$$

$$\therefore (1+i \tan \theta)^4 + (1-i \tan \theta)^4$$

$$= \frac{1}{\cos^4 \theta} \left[\cos^4 \theta + i 4 \cos^3 \theta \sin \theta - 6 \cos^2 \theta \sin^2 \theta - 4 \cos \theta \sin^3 \theta + \sin^4 \theta \right]$$

$$= \frac{2 \cos 4\theta}{\cos^4 \theta} \quad [3]$$

(ii)

$$z = i \tan \theta$$

$$(1+z)^4 + (1-z)^4 = 0$$

$$\Rightarrow \frac{2 \cos 4\theta}{\cos^4 \theta} = 0$$

note $\cos \theta \neq 0$

$$\therefore \cos 4\theta = 0$$

$$4\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$$

$$\theta = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}$$

2nd Quad = - (1st Quad)

$$\theta = \frac{\pi}{8}, \frac{3\pi}{8}, -\frac{3\pi}{8}, -\frac{\pi}{8}$$

 $\therefore$ roots of $z = i \tan \theta$

$$z = \pm i \tan \frac{\pi}{8}, \pm i \tan \frac{3\pi}{8}$$

[2]

(iii)

$$(1+z)^4 + (1-z)^4 = 0$$

$$\begin{array}{r} 121 \\ 1331 \\ 14641 \end{array}$$

$$1 + 4z + 6z^2 + 4z^3 + z^4$$

$$1 - 4z + 6z^2 - 4z^3 + z^4 = 0$$

$$2(1 + 6z^2 + z^4) = 0$$

$$1 + 6z^2 + z^4 = 0$$

$$\text{Let } m = z^2$$

$$m^2 + 6m + 1 = 0$$

$$(m+3)^2 = 8$$

$$m+3 = \pm 2\sqrt{2}$$

$$m = -3 \pm 2\sqrt{2}$$

$$z^2 = -3 \pm 2\sqrt{2}$$

$$\left(\pm i \tan \frac{\pi}{8} \right)^2 = -3 \pm 2\sqrt{2}$$

$$\text{or } \left(\pm i \tan \frac{3\pi}{8} \right)^2 = -3 \pm 2\sqrt{2}$$

$$-\tan^2 \frac{\pi}{8} = -3 \pm 2\sqrt{2}$$

$$-\tan^2 \frac{3\pi}{8} = -3 \pm 2\sqrt{2}$$

$$\tan^2 \frac{\pi}{8} = 3 \pm 2\sqrt{2} = \tan^2 \frac{3\pi}{8}$$

$$\tan \frac{\pi}{8} < \tan \frac{2\pi}{8} < \tan \frac{3\pi}{8}$$

$$\Downarrow$$

$$\tan^2 \frac{3\pi}{8} = 3 + 2\sqrt{2}$$

[2]

$$\frac{2c}{p(x)} = x^4 - 4x^3 - 3x^2 + 50x - 52$$

(i)

has a zero at $x = 3 - 2i$ $\therefore$ also at $x = 3 + 2i$ $\therefore (x - (3 - 2i))(x - (3 + 2i))$ is a factor.

$$(x^2 - 3 + 2i)(x^2 - 3 - 2i)$$

$$(x-3)^2 - 4i^2 = (x-3)^2 + 4$$

$$= x^2 - 6x + 13$$

$$P(x) = (x^2 - 6x + 13) Q(x)$$

either do synthetic division

$$\begin{array}{r} x^2 + 2x - 4 \\ x^2 - 6x + 13 \overline{) x^4 - 4x^3 - 3x^2 + 50x - 52} \\ \underline{x^4 - 6x^3 + 13x^2} \\ 2x^3 - 16x^2 + 50x \\ \underline{2x^3 - 12x^2 + 16x} \\ -4x^2 + 24x - 52 \\ \underline{-4x^2 + 24x - 52} \\ 0 \end{array}$$

OR
say $P(x) = (x^2 - 6x + 13)(x^2 + ax - 4)$
note needed to give -52

LHS of $P(x)$ coeff of $x^3 = -4$

RHS coeff of $x^3 = a - 6 \quad \therefore a = 2$

So $P(x) = (x^2 - 6x + 13)(x^2 + 2x - 4)$ [3]

(ii) over the complex field

$$\begin{aligned} x^2 + 2x - 4 &= 0 \\ x^2 + 2x &= 4 \\ x^2 + 2x + 1 &= 5 \\ (x+1)^2 &= 5 \\ x+1 &= \pm\sqrt{5} \\ x &= -1 \pm \sqrt{5} \end{aligned}$$

hence $(x+1+\sqrt{5})(x+1-\sqrt{5})$

$$P(x) = (x-3+2i)(x-3-2i)(x+1+\sqrt{5})(x+1-\sqrt{5})$$

[2]

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Question 3

a) (i) $\frac{x^2}{16} - \frac{y^2}{9} = 1$ hyperbola
 $a=4, b=3$

$$e^2 = 1 + \frac{b^2}{a^2} = 1 + \frac{3^2}{4^2} = \frac{25}{16}$$

$$e > 0 \quad e = \frac{5}{4} \quad [1]$$

(ii) foci $(\pm ae, 0)$

$$S_1 \text{ is } (4 \cdot \frac{5}{4}, 0) = (5, 0)$$

$$S_2 \text{ is } (-4 \cdot \frac{5}{4}, 0) = (-5, 0) \quad [2]$$

(iii) Directrices $x = \pm \frac{a}{e}$

$$x = \pm 4 / \frac{5}{4} = \pm \frac{16}{5} = \pm 3\frac{1}{5}$$

[2]

(iv) $\frac{x^2}{16} - \frac{y^2}{9} = 0$

$$\frac{2x}{16} - \frac{2y}{9} \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{2x}{16} \times \frac{9}{2y} = \frac{9x}{16y}$$

at $(5, 2\frac{1}{4}) = (5, \frac{9}{4})$

$$\text{slope of tangent} = \frac{9 \times 5}{16 \times \frac{9}{4}} = \frac{5}{4}$$

eqn of tangent:

$$y - y_1 = m(x - x_1)$$

$$y - \frac{9}{4} = \frac{5}{4}(x - 5)$$

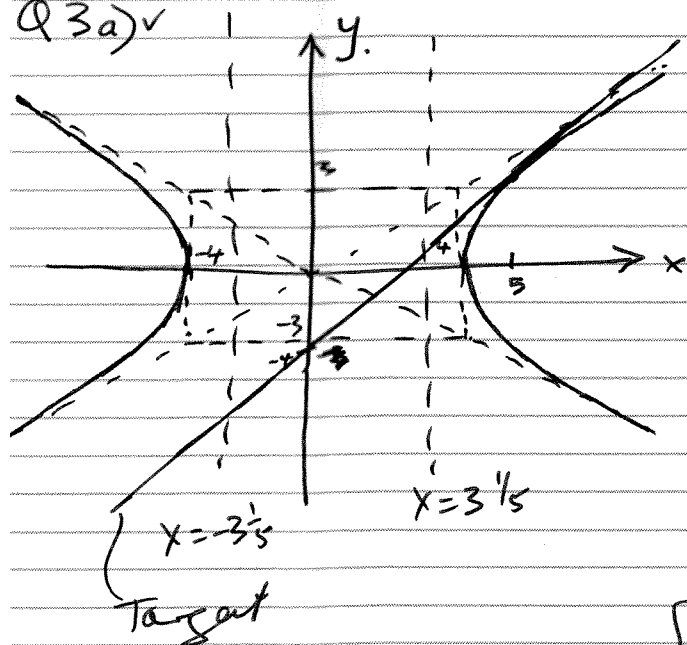
$$4y - 9 = 5x - 25$$

$$4y = 5x - 16$$

$$5x - 4y - 16 = 0 \text{ or } y = \frac{5}{4}x - 4$$

[2]

Q3a)✓



[2]

b) $\frac{x^2}{k-19} + \frac{y^2}{3-y} = 1$

note

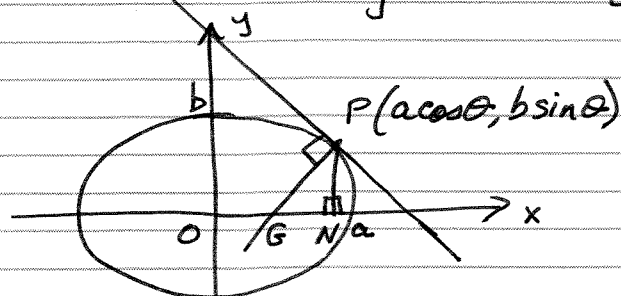
$$k-19 > 0 \Rightarrow k > 19$$

$$3-y > 0 \Rightarrow 3 > y$$

it is not possible $k > 19$
 $y < 3$

[1]

c)



(i) to show.

$$OG = e^2 \cdot ON$$

$$x = a \cos \theta \Rightarrow \frac{dx}{d\theta} = -a \sin \theta$$

$$y = b \sin \theta \Rightarrow \frac{dy}{d\theta} = b \cos \theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx} = \frac{b \cos \theta}{-a \sin \theta}$$

$$m_{\text{Tangent}} = -\frac{b \cos \theta}{a \sin \theta}$$

$$m_{\text{Normal}} = \frac{a \sin \theta}{b \cos \theta}$$

∴ eqn of PG

$$y - b \sin \theta = \frac{a \sin \theta}{b \cos \theta} (x - a \cos \theta)$$

to find G, let $y = 0$

$$-b^2 \sin \theta \cos \theta = a \sin \theta x - a^2 \sin \theta \cos \theta$$

$$(a^2 - b^2) \sin \theta \cos \theta = a \sin \theta \cdot x$$

$$x = \left[\frac{a^2 - b^2}{a} \right] \cos \theta$$

$$\therefore OG = \left[\frac{a^2 - b^2}{a} \right] \cos \theta$$

$$N \text{ is } (a \cos \theta, 0)$$

$$\therefore ON = a \cos \theta$$

$$OG = \left[\frac{a^2 - b^2}{a^2} \right] a \cos \theta$$

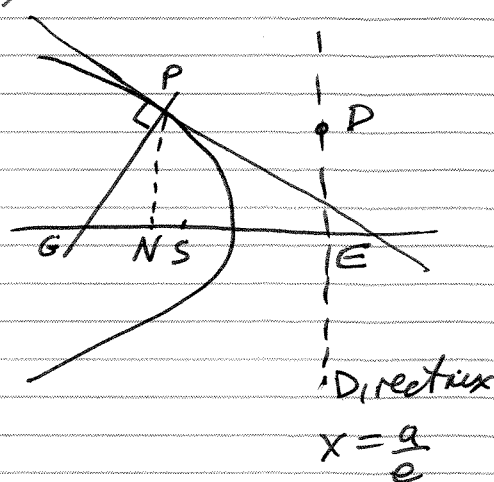
$$= \left[1 - \frac{b^2}{a^2} \right] a \cos \theta$$

$$= e^2 \cdot a \cos \theta$$

$$= e^2 \cdot ON$$

[3]

(ii)



Q3cii continue

by definition

$$\frac{P_s}{P_D} = e \quad \text{--- (a)}$$

[To show $SG = ESP$]

$$SG = OS - OG$$

$$= ae - \left(\frac{a^2 - b^2}{a} \right) \cos \theta$$

$$SP = e.PD \quad \text{from (a)}$$

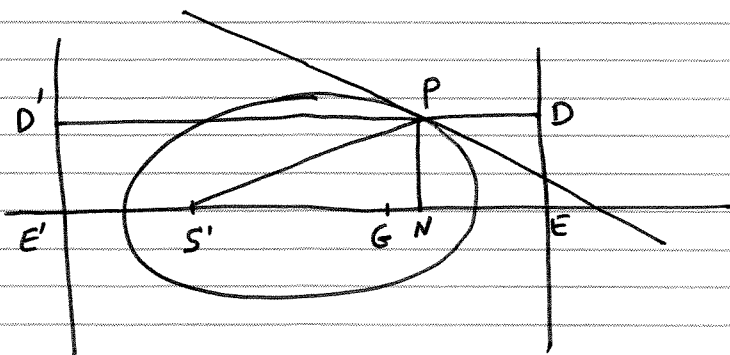
$$= e \cdot NE$$

$$= e \left[\frac{a}{e} - a \cos \theta \right]$$

$$= a - a \cos \theta$$

$$\begin{aligned}\therefore e. SP &= ae - ae^2 \cos \theta \\ &= ae - a \left(1 - \frac{b^2}{a^2} \right) \cos \theta \\ &= ae - \left(\frac{a^2 - b^2}{a} \right) \cos \theta\end{aligned}$$

$$eSP = SG$$



E' is $(-\frac{a}{e}, 0)$

$$\frac{S'P}{P'P} = e \quad \therefore S'P = e D'P$$

$$\begin{aligned} e \cdot S'P &= e^2 D'P = e^2 \cdot NE' \\ &= e^2 \left(a \cos \theta + \frac{a}{e} \right) \\ &= ae^2 \cos \theta + ae \end{aligned}$$

$$= \frac{a(a^2 - b^2)}{a^2} \cos \theta + ae$$

$$= \left(\frac{a^2 - b^2}{a} \right) \cos \alpha + a e$$

$$= S'G$$

$$\therefore SG = e \cdot SP$$

allow: [3]

Q3d

(1) $xy = c^2$

(1)

$$P(c_p, \frac{c}{p}) \quad Q(c_q, \frac{c}{q})$$

$$y = \frac{c^2}{x} \quad \frac{dy}{dx} = -\frac{c^2}{x^2}$$

at Q $\frac{dy}{dx} = -\frac{c^2}{c^2 y^2} = -\frac{1}{y^2}$

eqn of Tangent at Q:

$$y - \frac{c}{g} = -\frac{1}{g^2}(x - cg)$$

cuts axis at $N(cp, 0)$

$$0 - \frac{c}{q} = -\frac{1}{q^2}(cp - cq)$$

$$Cq = \cancel{Cp} - Cq$$

$$2Cg = Cp$$

$$p = 2q$$

[2]

(ii) M is midpoint of PQ.

$$x = \frac{c}{2}(p+q)$$

$$x = \frac{c}{2}(3q)$$

$$\Rightarrow \frac{2x}{3c} = 9$$

Q3d(ii)

$$y = \frac{c}{2} \left(\frac{1}{p} + \frac{1}{q} \right)$$

$$= \frac{c}{2} \left(\frac{1}{29} + \frac{1}{9} \right)$$

$$= \frac{c}{2} \left(\frac{3}{29} \right)$$

$$y = \frac{3c}{49} \Rightarrow q = \frac{3c}{49}$$

$$\frac{2x}{3c} = \frac{3c}{49}$$

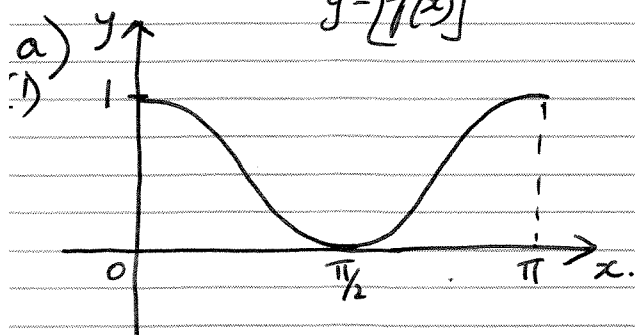
$$8xy = 9c^2$$

$$8xy = (3c)^2$$

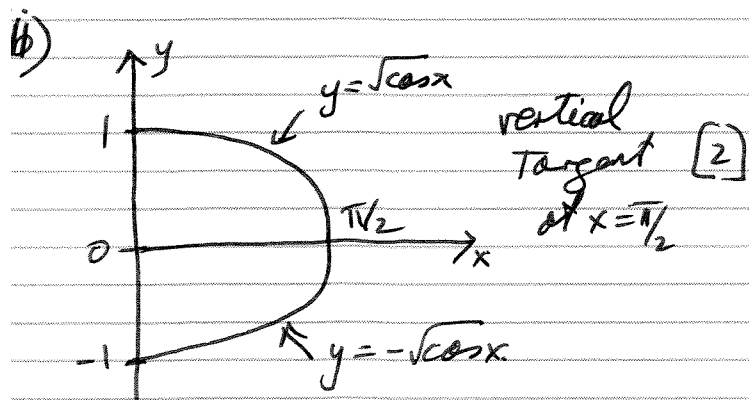
which is a rectangular hyperbola [2]

Question 4

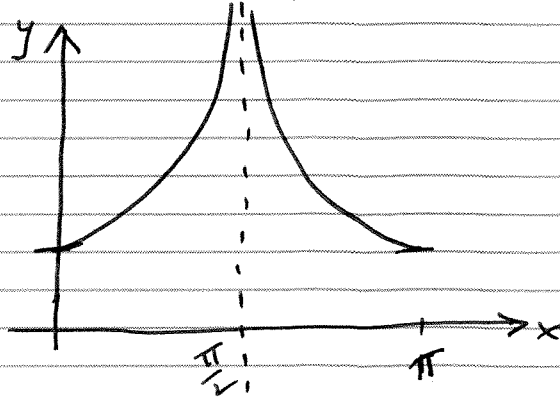
$$y = [f(x)]^2$$



note tangent is horizontal at $x=0, \pi/2, \pi$ [2]



(iii) $y = \frac{1}{|\cos x|}$



discontinuous at $x = \frac{\pi}{2}$
horizontal tangent at $y = 1$

b)(i)

$$\frac{5x}{(x+1)(x-2)} = \frac{a}{x+1} + \frac{b}{x-2}$$

$$= \frac{a(x-2) + b(x+1)}{(x+1)(x-2)}$$

$$= \frac{ax - 2a + bx + b}{(x+1)(x-2)}$$

$$= \frac{(a+b)x + (b-2a)}{(x+1)(x-2)}$$

equate coefficients

$$\left. \begin{aligned} a+b &= 5 \\ b-2a &= 0 \end{aligned} \right\} \text{ solve}$$

subtract $3a = 5 \quad a = \frac{5}{3}$

$$b = \frac{10}{3}$$

$$\frac{5x}{(x+1)(x-2)} = \frac{5}{3(x+1)} + \frac{10}{3(x-2)} \quad [2]$$

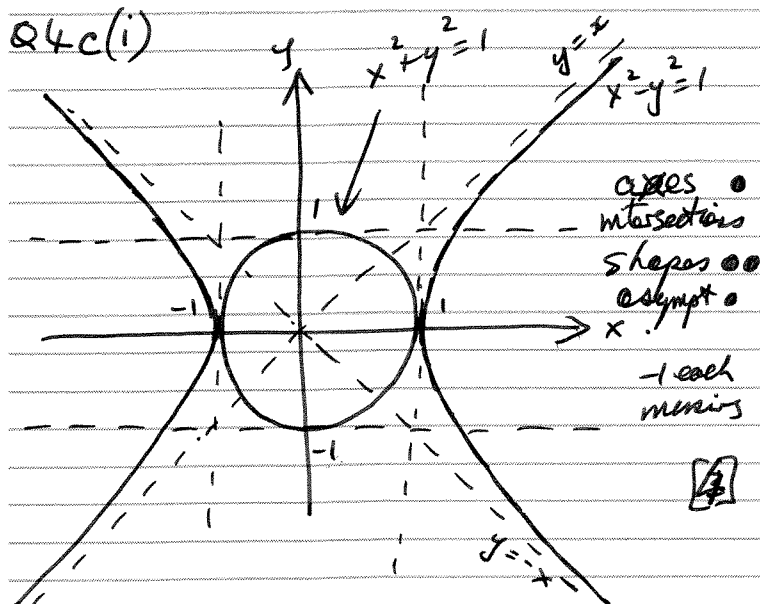
(ii)

$$I = \int_3^4 \frac{5x}{(x+1)(x-2)} dx$$

$$\begin{aligned}
 I &= \int_3^4 \frac{5}{3(x+1)} dx + \int_3^4 \frac{10}{3(x-2)} dx \\
 &= \frac{5}{3} \left[\ln(x+1) \right]_3^4 + \frac{10}{3} \left[\ln(x-2) \right]_3^4 \\
 &= \frac{5}{3} \left[\ln 5 - \ln 4 \right] + \frac{10}{3} \left[\ln 2 - \ln 1 \right] \\
 &= \frac{5}{3} \left[\ln 5 - \ln 4 + 2 \ln 2 \right] \\
 &= \frac{5}{3} \ln 5.
 \end{aligned}$$

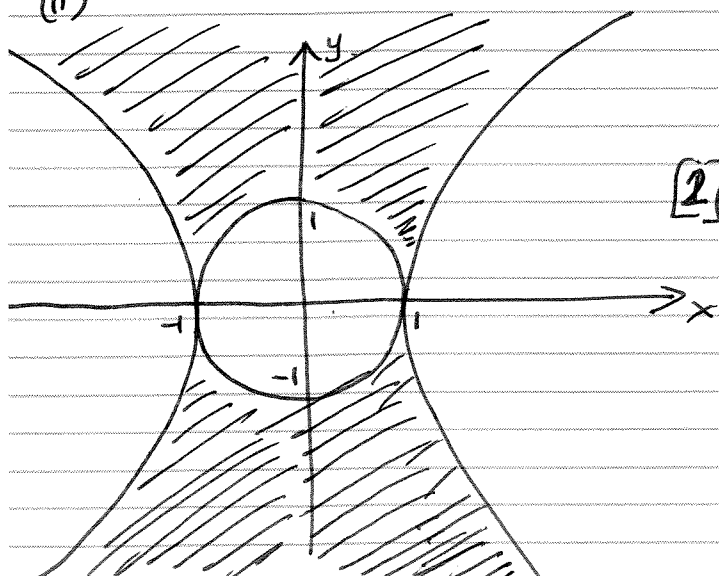
$$[2.682396 \approx 2.68 (2dp)]$$

Q4c(i)



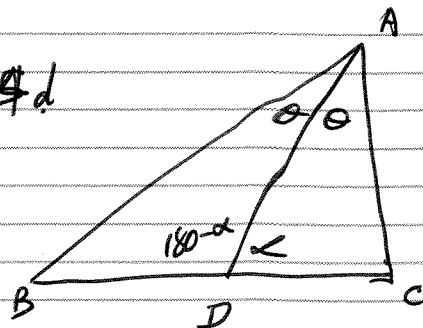
[4]

(ii)



[2]

Q4d



in $\triangle ABD$ $\frac{AB}{\sin D} = \frac{BD}{\sin \theta} \Rightarrow \sin(180-\alpha) = \frac{AB \sin \theta}{BD}$

in $\triangle ADC$ $\frac{\sin \alpha}{AC} = \frac{\sin \theta}{DC} \therefore \sin \alpha = \frac{AC \sin \theta}{DC}$

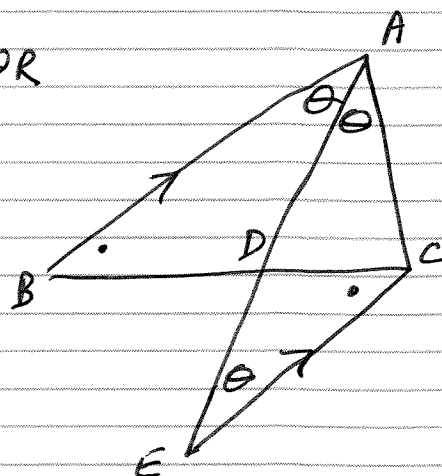
but $\sin(180-\alpha) = \sin \alpha$

$$\frac{AB \sin \theta}{BD} = \frac{AC \sin \theta}{DC}$$

$$\frac{AB}{BD} = \frac{AC}{DC}$$

[3]

OR



produce AD to E such that $BA \parallel CE$

Alternate angles equal as marked

$$\therefore \triangle ABD \parallel \triangle ECD \text{ (AAA)}$$

$$\therefore \frac{AB}{EC} = \frac{BD}{DC}$$

but $\triangle ACE$ is isosceles $\therefore CE = AC$

$$\frac{AB}{AC} = \frac{BD}{DC}$$

$$\therefore \frac{AB}{BD} = \frac{AC}{DC}$$

